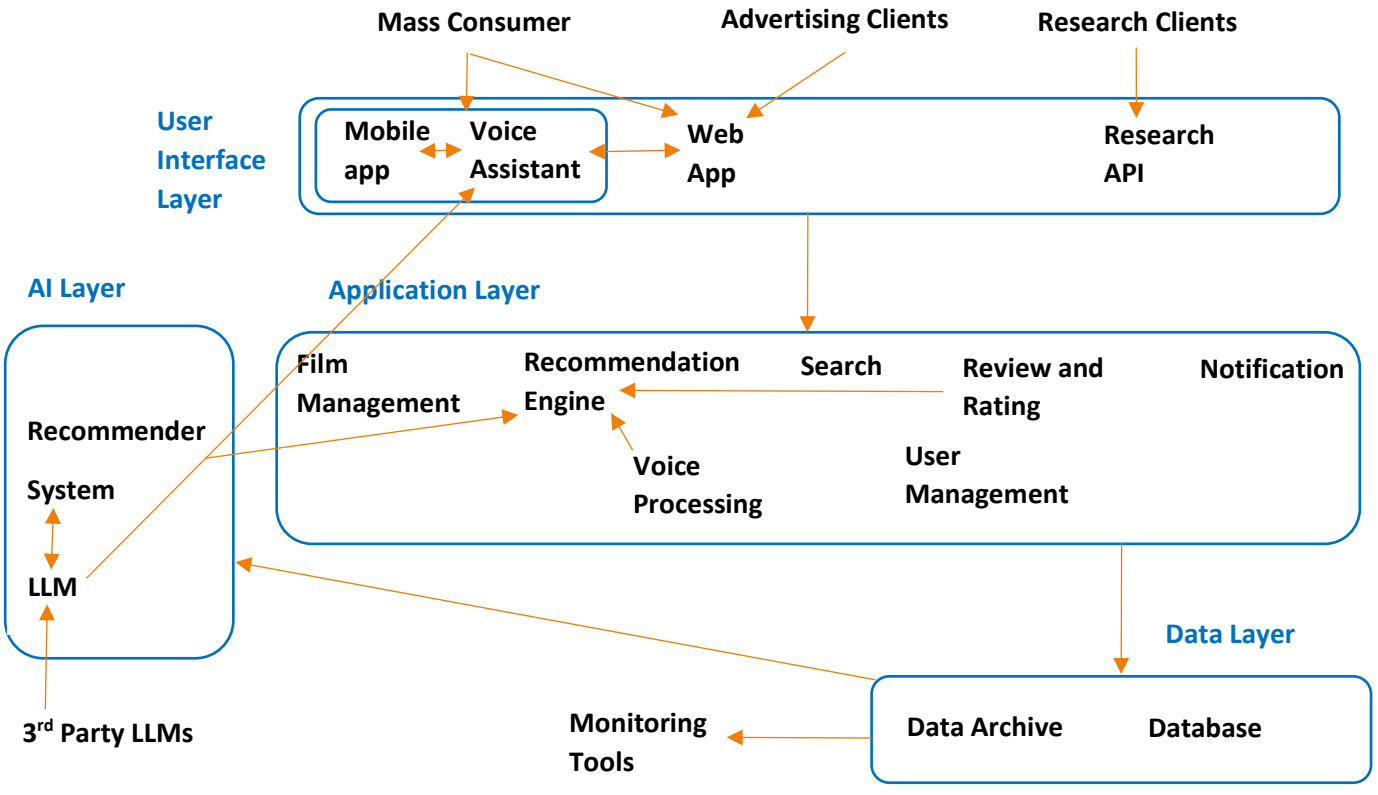
# AI Systems Engineering Assignment – Final Assignment.

## Introduction

The overall vision for the system is an online film recommendation database distinguished by a voice-enabled AI assistant – a film nerd friend in your pocket – that speaks to users in a friendly way to help them figure out which film to watch, and helps them find the right provider. The following assignment helps to define the design of the overall system – referred to as the "film recommender"

## System Architecture – Diagram and Explanation



**User Interface Layer.** The User interface Layer deals with how the film recommender interacts with its three types of clients: the mass consumer market; advertisers; and parties interested in using the data collected by the film recommender for research.

*Web App*. A web app must be developed to accommodate both mass consumer and advertising clients. The main functional requirement of the web app is therefore that of presenting two different user interfaces, one to mass consumers and the other for advertising clients so that they can monitor the performance of their paid-for advertisement. Since the unique selling point of the film recommender is the inclusion of a voice assistant, the web app must include voice assistant functionality for consistent branding, since this is the unique selling point of the film recommender. However, the web app will be more optimised for traditional film search and recommendation, integrating with those elements of the application layer, since the voice assistant is the more salient feature for the mobile app – the 'film nerd friend in your *pocket.*' The non-functional requirements for the web app are focused on seamless usability whilst integrating advertisement and links to third-party providers of media. Other non-functional requirements include rapid load times, adequate and compliant data protection, reliability in terms of minimising outages, compatibility with a wide range of browsers, and an easily maintainable code construction that makes scalability to new users or new features relatively simple.

*Mobile App*. The main functional requirements of the mobile app centre around constructing a user interface that doubles down on voice assistant interaction. The default user interface should prompt a conversation as soon as the app is accessed, and display film recommendations as the conversation progresses. Only if the user verbally requests for a traditional interface should it revert to a backup interface with lists of recommended film by category (such as trending, adventure, comedy, etc.) and a search bar. Effective implementation of all elements of the application layer is therefore also a functional requirement. The option to return to a conversational recommendation system must remain easily accessible even when the mobile app is in the traditional interface. The main non-functional requirements of the web app are similar to those of the web app, revolving around seamless user experience whilst enabling monetisation via the display of adverts and links to third-party film providers. Other non-functional requirements include rapid load times, adequate and compliant data protection, reliability in terms of minimising outages, compatibility with a wide range of phone operating systems, and an easily maintainable code construction that makes scalability to new users or new features relatively simple.

*Voice assistant.* The voice assistant must be more proactive than other voice assistants, and ready to ask unprompted questions in order to gather enough information that it can corroborate with the recommender system to display appropriate film recommendations. The voice assistant must also be able to conversationally enable other features of the film recommender, such as allowing users to rate films they have already seen or even give

feedback on the film recommender itself. In terms of non-functional requirements, the voice assistant must speak in an engaging and friendly manner.

*Research API.* The research API must allow for API key generation with different levels of price differentiation, so that concessional institutions (such as university students) can access the API for free whereas commercial clients can only receive API access with a subscription. Furthermore, paid-for APIs should have an enhanced user experience, and multiple levels of access, whereas concessional APIs should only remain basic in nature to save organisational resources. Advanced SQL-like querying will allow for complex data retrieval. The most important non-functional requirement is compliance with data protection laws, since this feature is the most prominent compliance risk vector.

**Application Layer.** The application layer holds the backend features that support the user interface layer whilst connecting with the database and AI systems.

*Film Management*. Film management concerns itself with managing the film database itself. Namely, CRUD (Create, Read, Update, Delete) operations are controlled here. Create in this context refers to the act of adding films to the database. Read refers to retrieval of a film from a database. Update refers to modifying the details of a film in the database, e.g. run-time if an extended version of the film is used instead. Delete refers to the elimination of films from the database.

*Recommendation Engine*. The recommendation engine coordinates the input of specific user data, the AI layer, and the database in order to actually generate film recommendations. This also regulates meta information about film recommendations, such as how many films should be recommended per query.

*Search.* The (traditional) search function is a backup to the voice assistant and allows searches via various criteria: title, genre, actor, and director. It also allows users to filter and sort the results, such as by run-time, or price from third-party providers.

*Review and Rating*. This facilitates the association of films with user ratings and reviews, ensuring this information is stored and that it feeds into the recommendation engine in order to improve recommendations.

*Notification*. The notification system facilitates notifications sent to users' phones and laptops, as well as marketing emails and texts. The notifications alert users of new releases or regular recommendations. Notifications are set up to be able to be personalized and completely disabled by the user.

*Voice Processing*. This concerns itself with coordinating and facilitating the elements that allow the AI-enabled voice assistant to function, as well as to be personalised. This includes keeping voice files of individual users for assisting in comprehension and the ability to choose the voice used by the assistant.

*User Management.* User management concerns itself with user registration and authentication. Authentication includes the option of two-factor authentication, though this is not a requirement. Registration includes Captchas.

**AI Layer.** The AI layer contains the artificial intelligence components of the film recommender, which centre around the large language model.

*LLM*. A large language model is the workhorse of the main AI-enabled feature of the film recommender – the voice assistant. This can either be an in-house model or a third-party model, allowing for flexibility in system design. The LLM directly controls the conversational aspect of the voice assistant, but its film recommender are further refined by the recommender system.

*Recommender System.* The job of the recommender system is to train and apply an appropriate set of AI models in order to incorporate user-specific information to improve the recommendations of films outputted to the user after interaction with the voice assistant. It is also responsible for the entirety of film recommendations when users input a more traditional search.

**Data layers.** The data layer is where the data that underpins the film recommender is stored.

*Database.* This stores user data, film data, reviews, and ratings. Utilizes both SQL and NoSQL databases for structured and unstructured data.

*Data Archive*. This aggregates historical data for analysis, freeing up the database itself of users that are no longer registered or films that are no longer in the database. Film and user management from the application layer, which regulate CRUD for films and users, integrate with the database and data archive for seamless transition from one to the other. The data archive can be used and, with modifications to protect data privacy, sold in the research API for research purposes.

*Monitoring tools*. Monitoring tools need to be set up to monitor the performance of the film recommender system as a whole. The use of both the data archive and current database is important to keep track of the performance of the film recommender over time, including looking at trends in user numbers and film capacity.

## RISKS

Incorrect predictions in the film recommender system can lead to several risks that ultimately affect the safety and commercial viability of the system. Below are some identified risks along with corresponding mitigation strategies at the system level:

**Risk 1: User Dissatisfaction**

User dissatisfaction at unsatisfactory film recommendations ultimately leads to disengagement, complaints, and bad review. Disengagement means fewer users, and therefore translates into less revenue from advertisement, third party content referrals, and data sales. Lesser revenue also ultimately means a lesser capacity to dedicate resources to maintaining and upgrading the film recommender. Lower data sales due to lower quality data from fewer disengaged users also add pressure to the revenue side, whilst complaints add pressure on the cost side.

**Mitigation Strategy**: **Continuous Learning and Feedback Loop**

*Description.* Learning from users comes from multiple sources: evaluation of the conversation that users have with the voice assistant; review and rating of recommended films that users have watched, and monitoring of the trends in terms of user retention. Analysis of conversations that users have with the voice assistant can be analysed to evaluate both when recommendations fail and why. Similarly, ratings and reviews can be used to track the performance of the system and evaluate why the user did or did not like a film, and whether or not it had previously been recommended by the system. Feeding these pieces of information back into the AI-enabled components can help update them to produce better predictions. Similarly, these and the trends produced by the monitoring tools can help inform management-level decisions to improve both the recommendations and concomitant features, such as how the recommendations are displayed.

*Extent of mitigation.* The greatest weakness of this strategy is in its partial reliance on human management skills in order to look honestly at any potential unfavourable trends and review the feedback systems being implemented. Denial, groupthink, and other cognitive biases can impede effective resolution of user dissatisfaction, although the strategy has the potential to fully address user dissatisfaction.

**Risk 2: Bias in recommendation**

Bias in the recommendation system can lead to unfair or skewed suggestions, impacting the diversity of films recommended and potentially alienating certain user groups, as well as potentially attracting negative press attention.

**Mitigation Strategy**: **Regular Audits and Diversity Algorithms**

*Description.* Conduct regular audits of the recommendation outputs to detect and correct biases. Implement algorithms that promote diversity in recommendations by ensuring a wide range of genres, directors, and cultural representations. However, disclose (the default) algorithm diversity adjustments and provide users with the option to disable the enhancement, since the inclusion of these can itself be controversial. The extent to which any diversity adjustment is deployed or obvious can be varied depending on geography, as diversity enhancement can itself be controversial.

*Extent of mitigation*: No strategy can eliminate all potential backlash both by proponents and opponents of enhanced diversity in algorithmic recommendations. By varying the extent of diversity enhancement and its advertisement based on political geography and retaining user customizer, backlash can be minimised. Continuous management supervision of the political environment is needed to maintain the effectiveness of this mitigation.

**Risk 3: Data Privacy Concerns**

Incorrect predictions based on sensitive data might raise privacy concerns among users, especially if they feel their personal preferences are being misused or overly scrutinized. The use of cookies or possible incorporation of third-party data on users is most likely to lead to such an error. Examples of other contexts in which such errors have caused offence (such as retailers' algorithms suggesting pregnancy-related products) invite us to be weary of insensitivity with data.

**Mitigation Strategy**: **Data Anonymization and Privacy Policies**

*Description.* Where relevant, anonymize user data to prevent the identification of individuals. Given that some personalised data can be useful for prediction purposes, a vetting process of data sources used to personalise recommendations must be put in place. Furthermore, Clearly communicate privacy policies and how user data is used, and ensure compliance with data protection regulations such as GDPR and CCPA.

*Extent of mitigation.* With black-box AI models, it is difficult to completely control how and what predictions will be based on, leaving some risk even with the mitigation strategy.

## DEPLOYMENT

**Stage 1: Public Cloud Model**

Starting with a public cloud model of deployment is appropriate for the early stages of company development. Starting with a software as a service, the lack of technical expertise from a startup technical team can be supplemented by a third party provider that hosts the majority of the components of the film recommender, including AI components. As in-house expertise scales up, the deployment model can transition to lower levels of abstraction within the public cloud model, moving to platform as a service and then infrastructure as a service. scalability at a lower cost will be more crucial as the company begins to grow and add to its in-house capabilities.

**Stage 2: Hybrid model**

As the company scales, data security and intellectual property will rise in importance, leading to the suitability of a hybrid model vs a purely public cloud model. Data and the AI components of

the film recommender can be hosted on the private cloud component, leaving other components to still be hosted on the public cloud to lessen the burden on the in-house team.

**Stage 3: Private Cloud Model**

A company that has grown sufficiently can then transition to a private cloud model. Not only does a private cloud model offer more protection of intellectual property and data, but for a large company it also allows for cost control.

## Relationship Between Goals

Telemetry for the film recommender emanates from implicit user interactions (such as clicks), explicit user feedback such as ratings and reviews, and system performance metrics, and user trends.

**Implicit user interactions**

*Clicks.* Clicks, or taps in case of a mobile app, can indicate interest in recommended films. The number of clicks that elaborate on the details of a film indicate interest, whereas clicks off to third-party providers are an even stronger indication of interest. Clicks can therefore be used to create metrics for the relevance of recommendations. For users receiving recommendations via the voice assistant, the metric could be number of rounds of recommendations before the user taps on a film, or taps on a film and taps to receive more information, or before a user taps on to a third party provider. An easier system-wide metric is number of clicks/taps over the total number of recommendations viewed, and a more refined version of this metric focuses on number of clicks off-platform.

*User interaction with voice assistant*. The interactions with the voice assistant require a greater level of analysis due to the complex nature of the data they contain. The transcripts of the interactions with the voice assistant should be run through natural language processing techniques such as opinion mining to draw associations between film recommendations and user feedback. Other than that, the time spent with the voice assistant before selecting a film can be used to evaluate performance, although this must be corroborated with other metrics since the association between time spent talking to the voice assistant and the performance of the system is unclear – more time spent with the voice assistant could indicate an engaging interaction or a frustrating one that does not find a desired film. Owing to the complex nature of this data, it is likely that the volume of data will be large enough to require careful treatment.

**Explicit user feedback**

Explicit user feedback is received via the ratings and reviews users make about films. Although users can leave ratings and reviews about any film on the database, user logs can indicate if the

user wrote a review or rated the film after its being recommended. Using ratings and opinion mining, the recommender can be scored. Sentiment analysis can be used to determine the overall sentiment of a review of a recommended film, allowing us to arrive at a net positive/negative count for film recommendations, normalised by the number of recommendations or total number of reviews. Similarly, with ratings, the recommendations of films can receive an average score from 1 to 5 starts from the post-recommendation ratings films receive.

**System performance metrics**

Not all telemetry is focused on the quality of recommendations themselves, and instead some should focus on system performance. System performance metrics include response times to produce recommendations, both from the use of a traditional search engine and the voice assistant. Short response times are key to creating a seamless user experience and therefore monitoring of response times is important information to improve the film recommender overall. Similarly, keeping track of error logs is a relatively data-light way to monitor system performance. Resource utilization statistics such CPU, memory, ad network usage statistics can also be used to enhance resource allocation.

**User Trends**

User trends were previously discussed when talking about the monitoring tools. By keeping a historical archive of users and films, the average length of user accounts can be monitored to help evaluate retention, with a relatively light data burden.